Reg. No.	R	A	2	3	1	1	0	0	3	0	1	0	4	2	2

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024
First and Second Semester

21CYB101J - CHEMISTRY
(For the candidates admitted from the academic year 2022-2023 onwards)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks)
Answer **ALL** Questions

	Marks	BL	CO	PO
1. The coordination number, oxidation number, number of electron in d-orbital and Number of unpaired electrons on complex $\left[Co(en)_2(NH_3)_2\right]^{2+}$ is _____respectively.	1	1	1	1
(A) 4,2,7,1 (B) 6,2,6,1 (C) 6,2,7,1 (D) 4,2,6,3				
2. The correct increasing order of splitting power of ligands according to the spectrochemical series is	1	2	1	1
(A) Cl<OH<CN (B) Cl<CN<OH (C) OH<Cl<CN (D) OH<CN<Cl				
3. The effective nuclear charge experienced is electron of Helium atom is	1	1	1	1
(A) 0.7 (B) 1.70 (C) 1.65 (D) 0.3				
4. Classification of acids and bases as HSAB was putforth by	1	1	1	1
(A) Lewis (B) Arrhenius (C) Pearson (D) Bronsted				
5. Internal energy does not include	1	1	2	1
(A) Energy of gravitational pull (B) Nuclear energy (C) Vibrational energy (D) Rotational energy				
6. The chemical often used in salt bridge is	1	1	2	1
(A) KOH (B) KCl (C) $KMnO_4$ (D) KBr				
7. According to the Pilling – Bedworth rule, greater the specific volume ratio	1	2	2	1
(A) More is the oxidation corrosion (B) Lesser is the oxidation corrosion (C) More is the Reduction corrosion (D) More the corrosive nature of the environment.				

8. Which of the following conditions are satisfied when cell reaction in the electro chemical cell is spontaneous? 1 2 2 1

(A) $\Delta G^\circ > 0$ (B) $E^\circ cell < 0$

(C) $E^\circ cell = 0$ (D) $\Delta G^\circ < 0$

9. How many optical isomers are possible in a compound with one chiral carbon? 1 2 3 1

(A) 3 (B) 2

(C) 4 (D) 5

10. In gauche conformation, the methyl groups are 1 3 3 1

(A) 90° apart (B) 60° apart

(C) 180° apart (D) 360° apart

11. When the nucleophile RO^- attacks the reaction, the resultant product will be 1 3 3 2

(A) ROH (B) RNHR

(C) RCN (D) ROR

12. The plane which divide the molecule into two equal parts so that each half is the mirror image of other half is called 1 2 3 1

(A) Ceter of symmetry (B) Axis of symmetry

(C) Plane of symmetry (D) Angle of symmetry

13. A polymer with an amide linkage is known as _____ 1 3 4 2

(A) Teflon (B) Terylene

(C) Nylon -6 ; 6 (D) Bakelite

14. Which of the following is NOT an example of thermoplastic? 1 1 4 1

(A) Polyvinyl chloride (B) Nylon

(C) Epoxy (D) Polysters

15. The number of repeating unit (n) in polymer molecule is called 1 1 4 1

(A) Degree of polymerization (B) Oligomer

(C) Heavy Polymer (D) Functionality

16. Most commonly used rubber vulcanization agent is 1 1 4 1

(A) Bromine (B) Platinum

(C) Sulphur (D) Alumina

17. _____ is the property of recovering original shape after the removal of deforming strain 1 1 5 1

(A) Elasticity (B) Rigidity modulus

(C) Youngs modulus (D) Bulk modulus

18. X-ray diffractometers are not used to identify the physical properties of which of the following. 1 3 5 1

(A) Liquids (B) Metals

(C) Solids (D) Polymeric materials

19.	The source for XPS is (A) Global source (B) Nernst glower (C) Mercury – arc (D) $A\ell k_{\alpha}$	1	3	5	1
20.	Longitudinal strength of fibre reinforced composite is mainly influenced by (A) Wear resistance (B) Elastic modulus (C) Fibre strength (D) Interface strength	1	2	5	1

PART – B (5 × 8 = 40 Marks)
Answer **ALL** Questions

		Marks	BL	CO	PO
21. a.i.	Describe the crystal field splitting of an octahedral complex with neat diagram.	5	3	1	1
ii.	Write a short note on Spectrochemical series.	3	2	1	1
	(OR)				
b.i.	Brief about any two isomerism exhibited by the coordination compounds with a suitable example for each.	4	3	1	1
ii.	Define and explain any two periodic properties.	4	2	1	1
22. a.i.	Comment on the statement "The entropy of the universe is increases".	4	3	2	1
ii.	Construct an electrochemical cell and give its representation.	4	1	2	1
	(OR)				
b.	Derive Nernst equation and give its application.	8	3	2	3
23. a.	Explain the various conformations of n-butane in detail with potential energy diagram.	8	3	3	1
	(OR)				
b.	Explain Cahn-Ingold prelog priority rules to determine R/S configuration on a chiral center taking an example.	8	3	3	1
24. a.	Write short notes on:				
i.	Vulcanization	2	2	4	2
ii.	Functionality	3	2	4	1
iii.	Tacticity	3	2	4	1
	(OR)				
b.	Brief on conducting polymers.	8	3	4	2
25. a.	Explain Bragg's law. Illustrate how its principle is applied to study the diffraction of X-ray by atoms in a crystalline structure with a neat sketch?	8	3	5	1
	(OR)				
b.i.	How the composite materials are classified?	4	1	5	1
ii.	Elaborate the stress-strain relationship of solids with a neat plot.	4	2	5	1

PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Question

		Marks	BL	CO	PO
26.i.	Draw a neat sketch of Pourbaix diagram and explain the zones and lines in it.	10	3	2	1
ii.	Determine the effective nuclear charge experienced by 4 p electron of Bromine (Atomic Number of Br = 35).	5	3	1	1
27.i.	Explain the principle and instrumentation of X-ray photo electron spectroscopy.	10	3	5	1
ii.	Compare and contrast SN^1 and SN^2 reaction.	5	3	3	2

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